



ENGG4800 – Project Management

Progress and Performance Measurement and Evaluation

Dr Wayes Tushar

Announcement

- ✓ ☒ **White paper marks** have been released. Peer assessment has been released as well.

$$\begin{aligned}\text{Individual student mark} &= \text{Team mark} \times \text{Peer assessment factor} \\ &= 80 \times 1.1 = 88\end{aligned}$$

- ☐ **Presentation schedule:** will be finalised and announced in Week 9.
 - Please indicate your availability by the end of this week (Week 8)
- ☐ **Lectures in Week 9 and Week 10 – Guest Lectures.** Please try to attend. Share with your friends.

Guest Lecture – Week 9

Guest Lecture: Project Management in Practice: Insights from a Remote Hardware Refresh Initiative



Mr Alexander Schuer
Project Manager
Origin Energy

- Over 17 years of industry experience across large-scale IT, operational technology and organisational transformation projects.
- Led several complex, multi-million dollar initiatives

Guest Lecture – Week 10

Guest Lecture: Intellectual Property Commercialisation



Dr Larry Weng

Director

Commercialisation Physical Sciences

UniQuest

- More than 20 years of experience in commercialising intellectual property
- Assisted over 100 startup founders in navigating the competitive grant application process

Plan for Rest of the Semester

Week 8 (This week)

- Practical 4
- Clarify any concerns with your practical problems

Week 9

- Guest Lecture 1
- Practice Practical Examination – During Class Time

Week 10

- Guest Lecture - 2
- Practical Examination (08 May 2026) – During Class Time

Plan for Rest of the Semester

Week 11

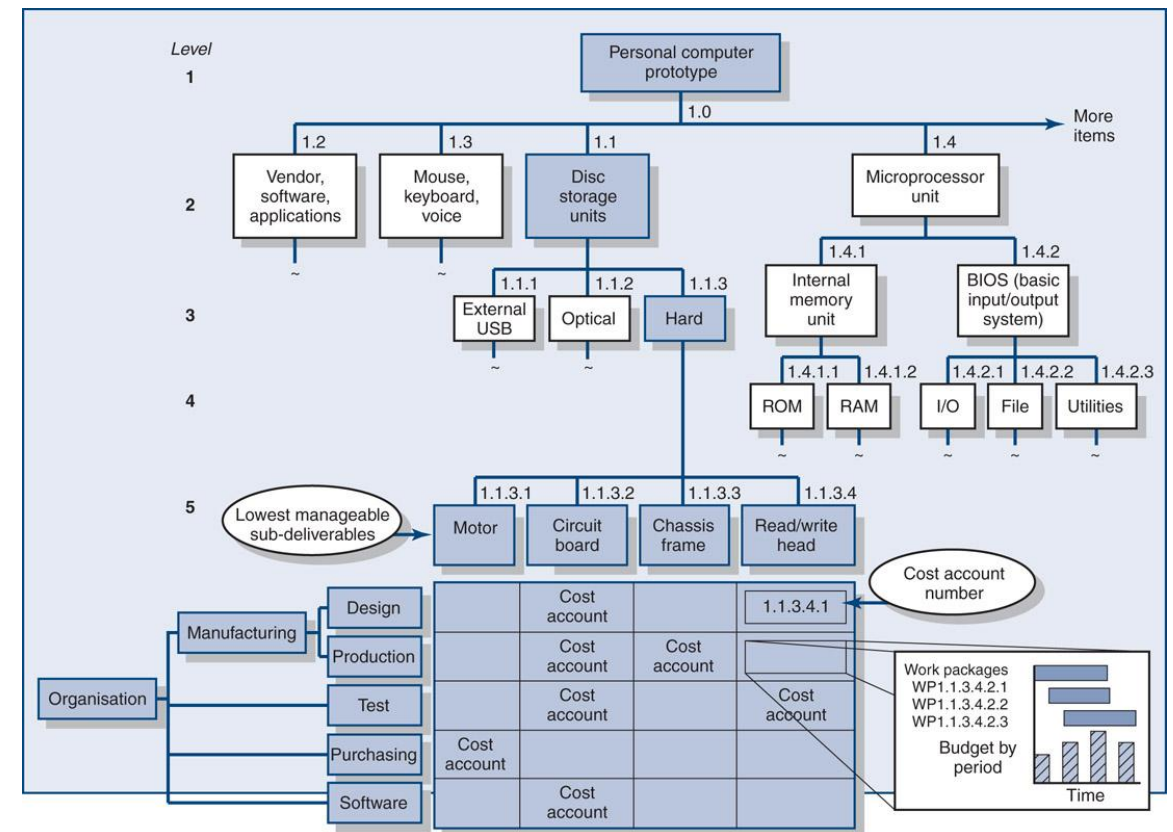
- No Lecture, No Tutorial
- Presentation (11, 12, 13 and 14 May 2026) – Scheduled time will be published on the BB
- Def Practical Examination (15 May 2026) – during the practical class
 - If the school approves Defer exam requests

Week 12

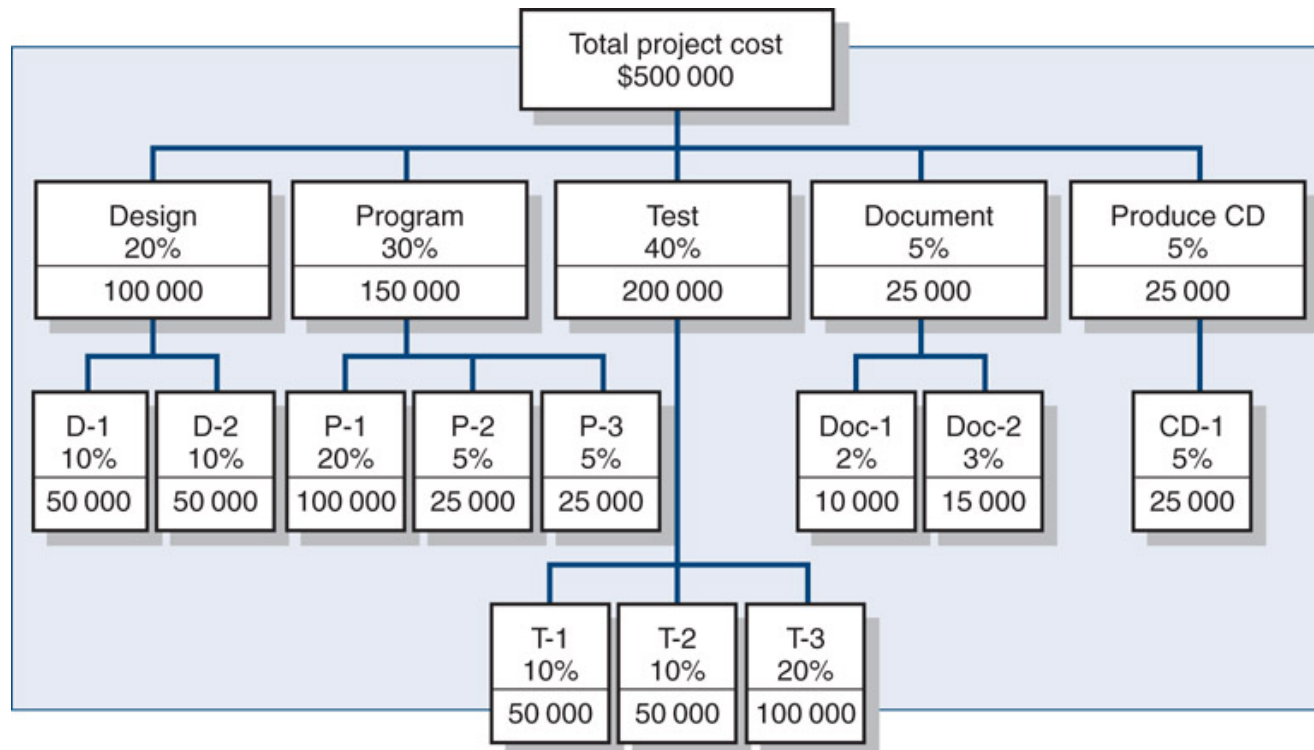
- Guest Lecture 3

Week 13

- Guest Lecture 4
- Final Project Proposal Due (27 May 2026)



Recap – what we learned so far?



Top-down estimates

Intended use

Feasibility/conceptual phase
Rough time/cost estimate
Fund requirements
Resource capacity planning

Preparation cost

1/10 to 3/10
of a per cent
of total project cost

Accuracy

Minus 20%
to plus 60%

Method

Consensus
Ratio
Apportionment
Function point
Learning curves

Bottom-up estimates

Intended use

Budgeting
Scheduling
Resource requirements
Fund timing

Preparation cost

3/10 of a per cent
to 1.0 per cent
of total project cost

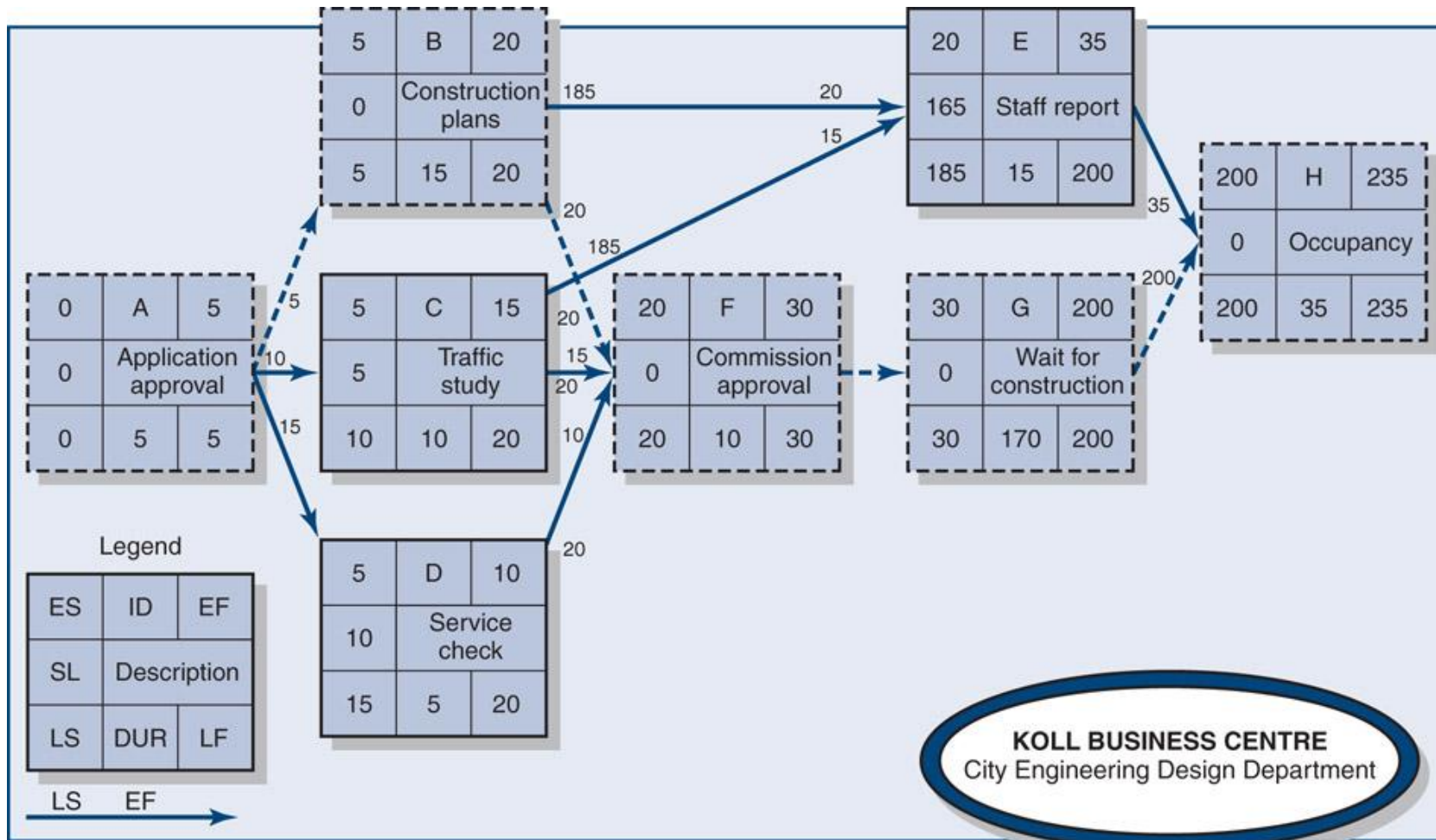
Accuracy

Minus 10%
to plus 30%

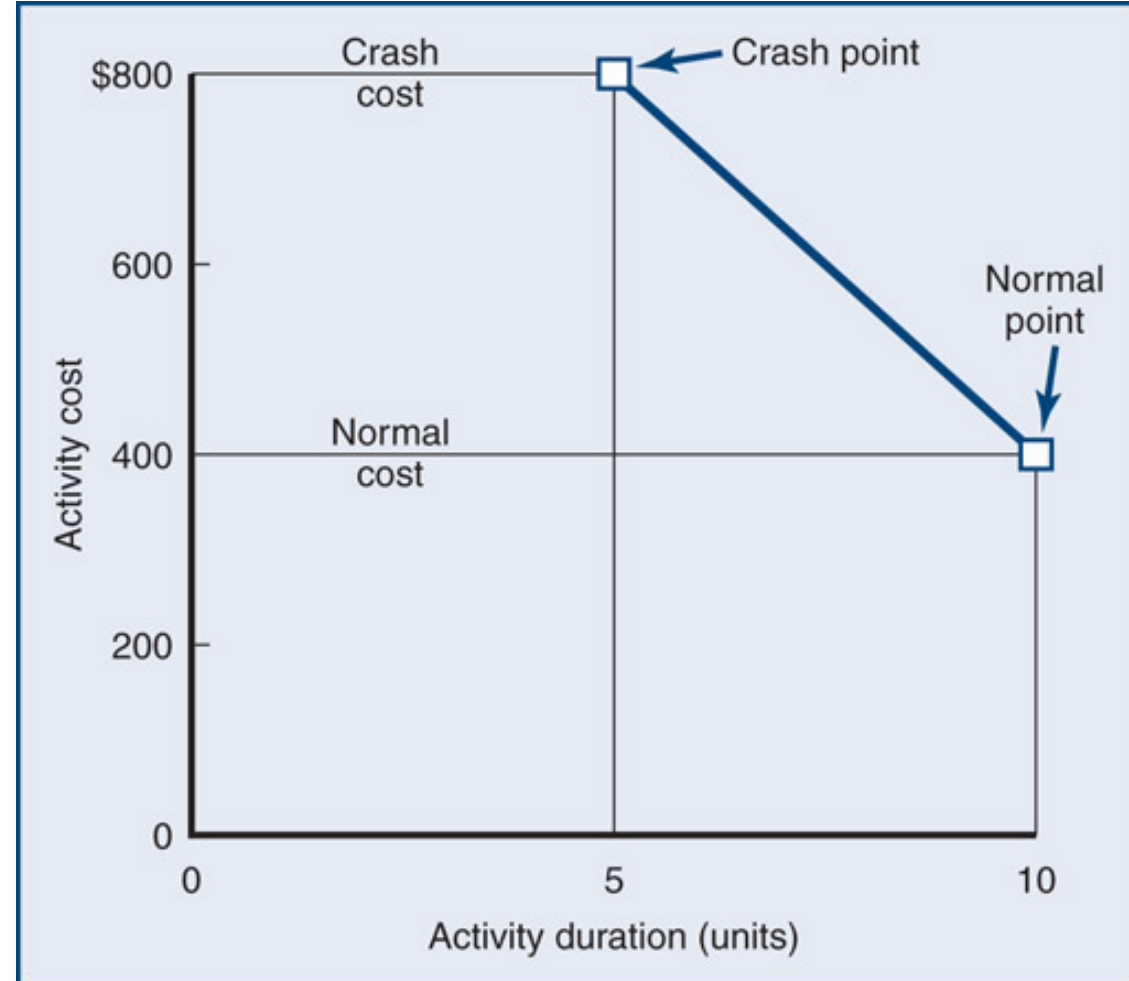
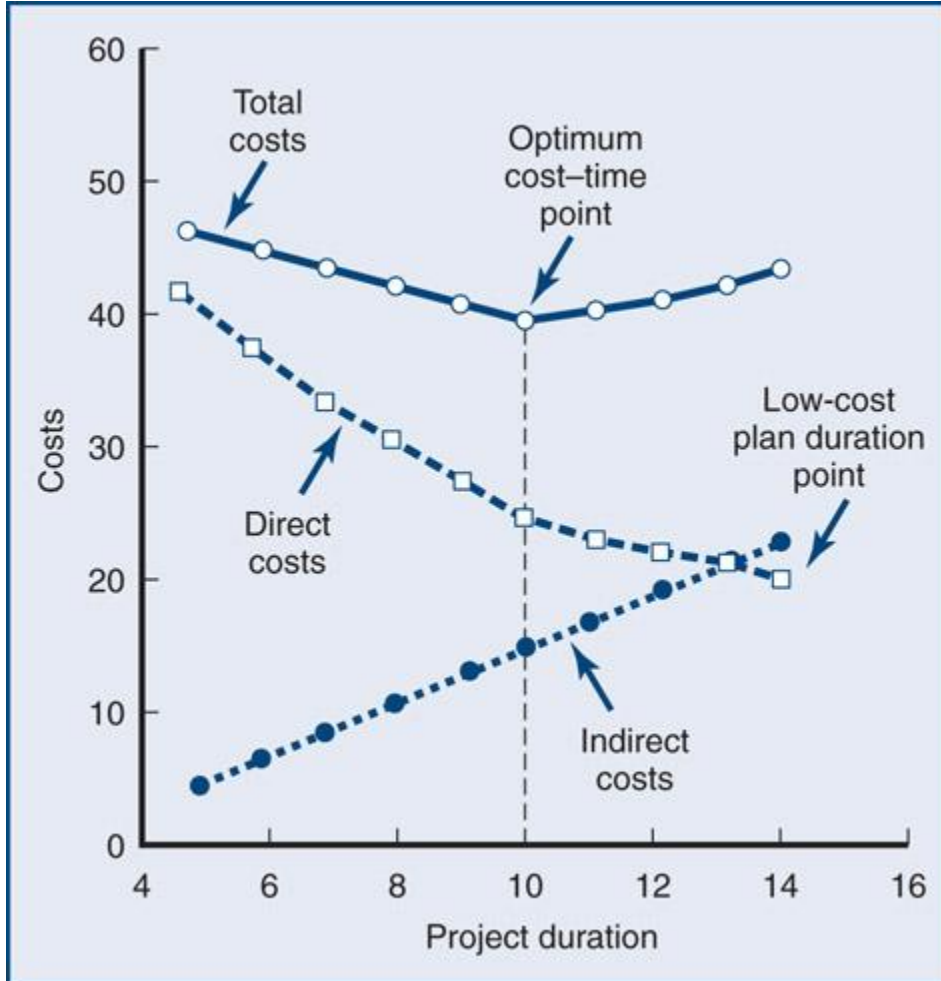
Method

Template
Parametric
WBS packages
Range estimates

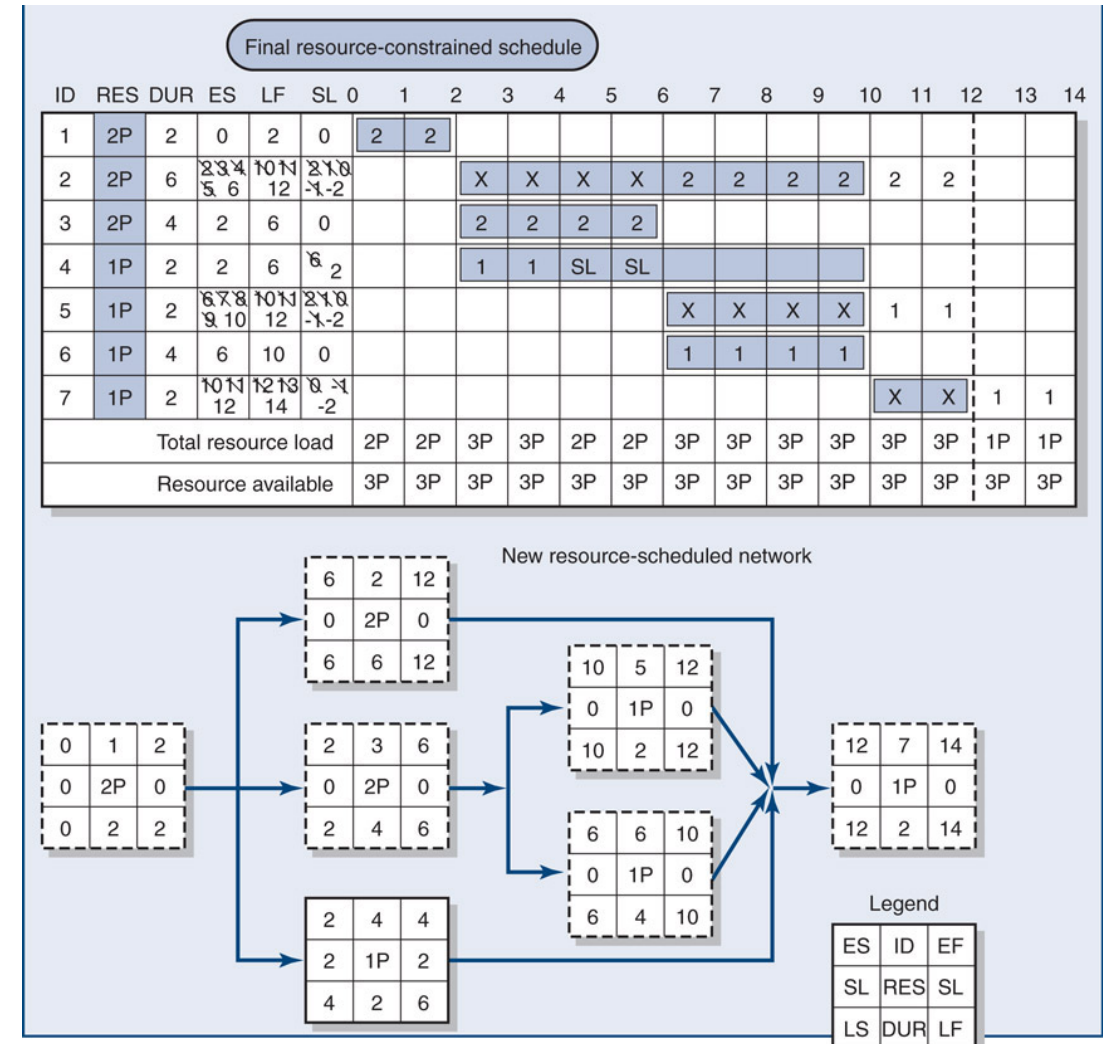
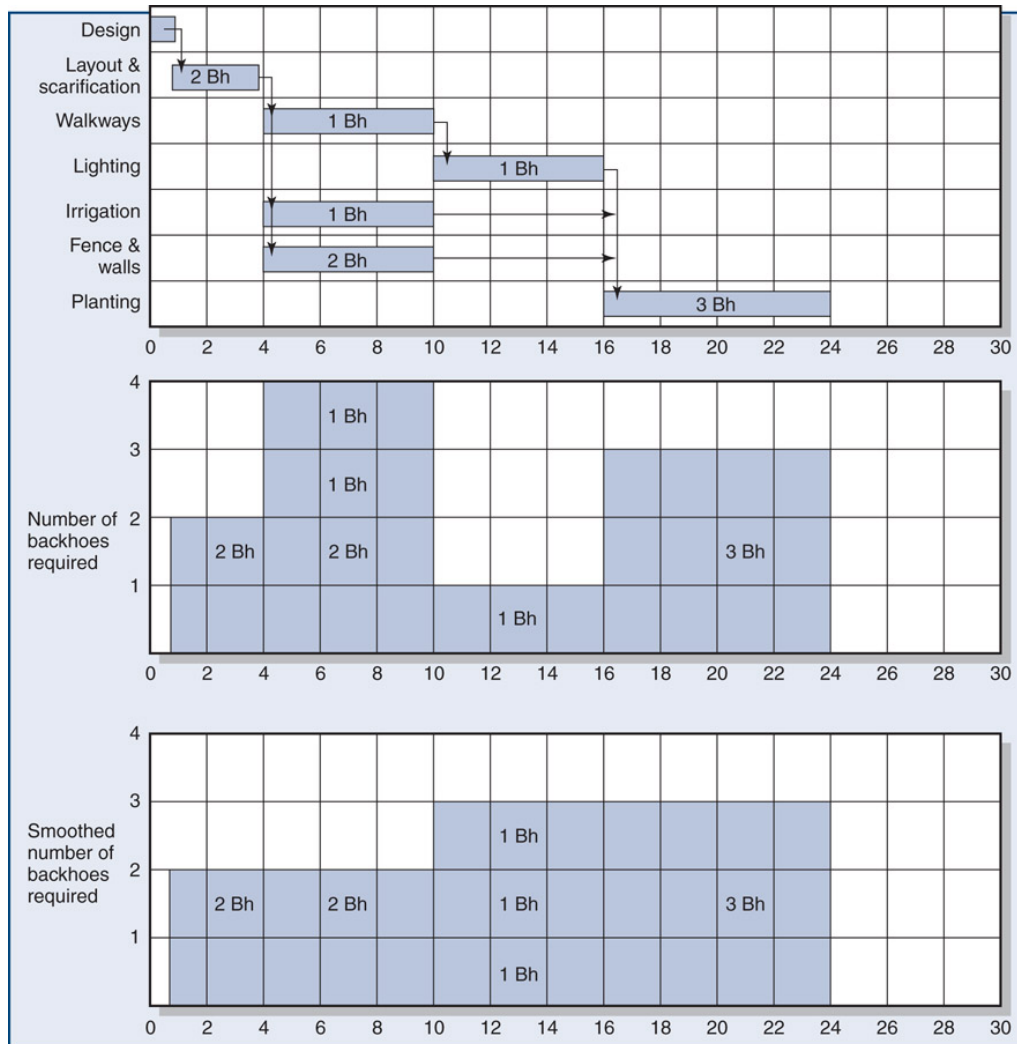
Recap – what we learned so far?



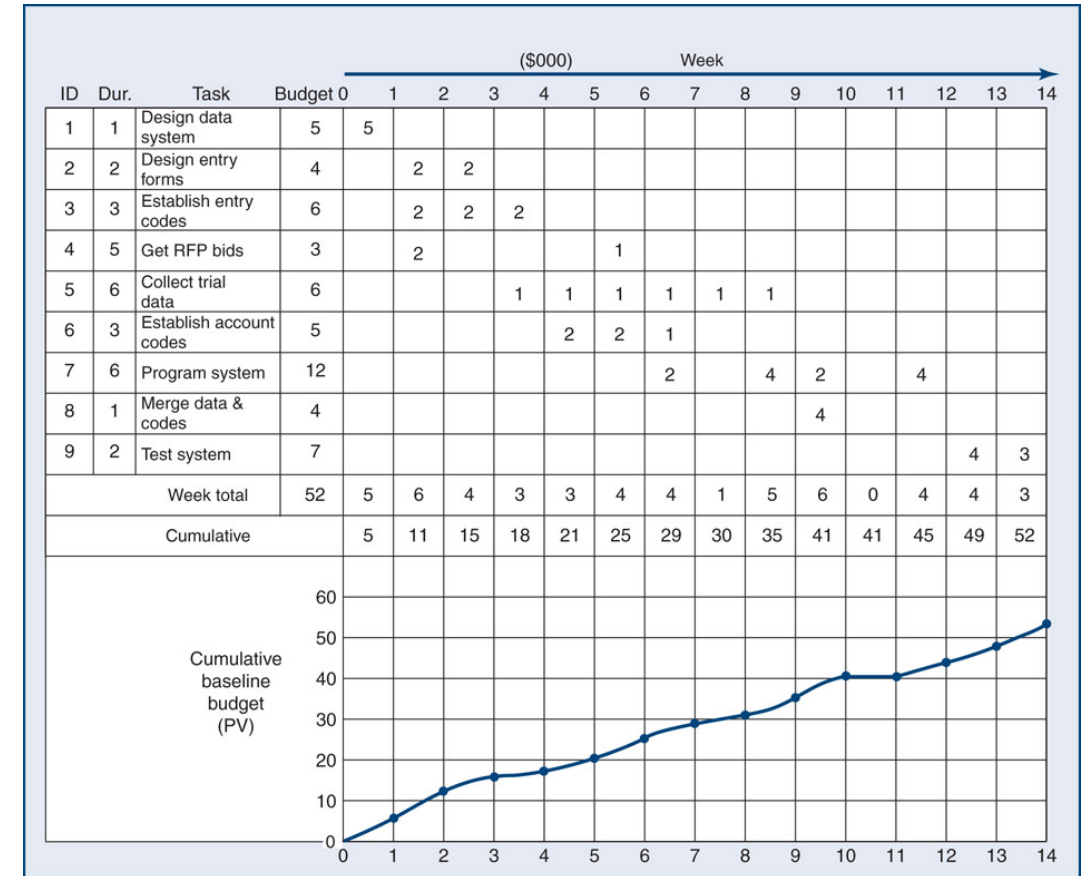
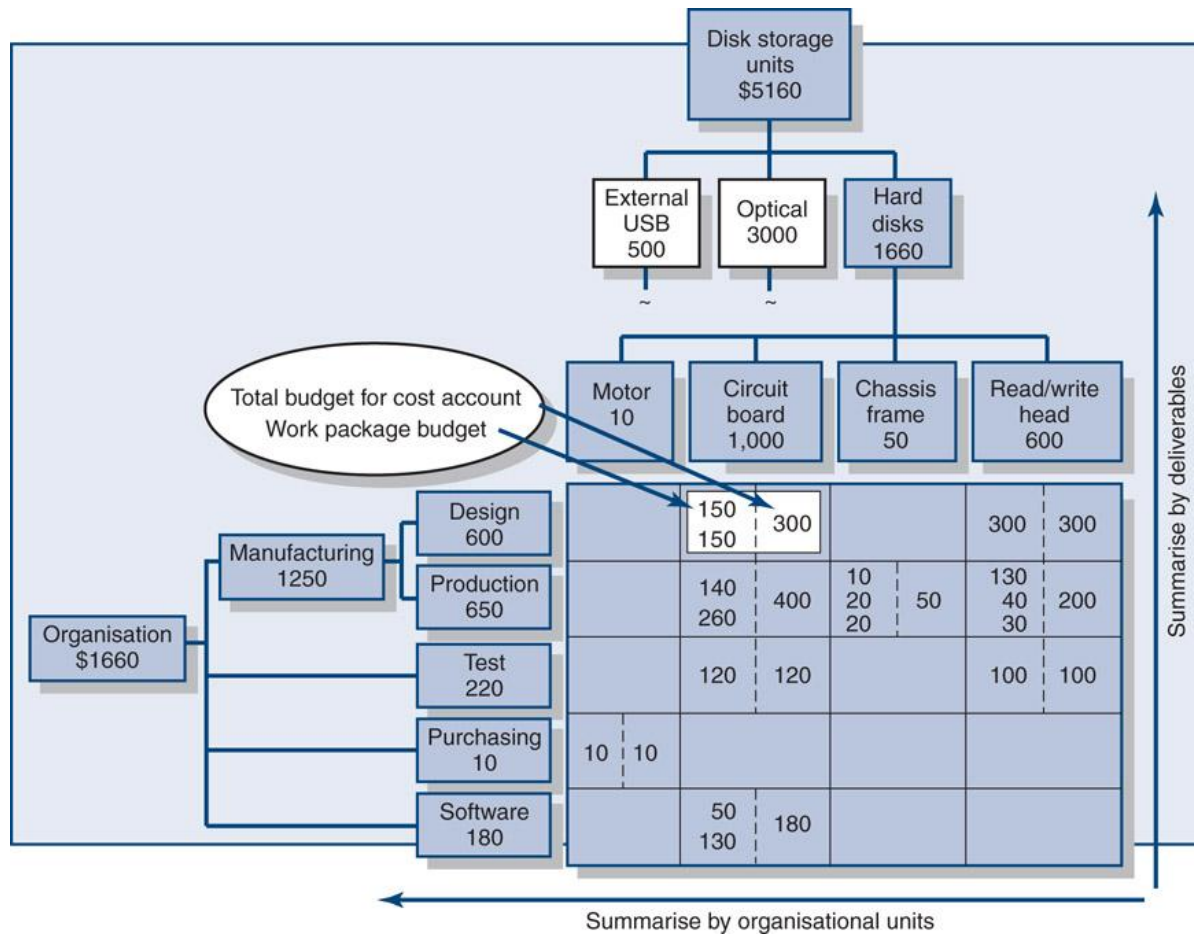
Recap – what we learned so far?



Recap – what we learned so far?



Recap – what we learned so far?



Learning objectives

- ❑ The project control process
- ❑ Monitoring time performance
- ❑ Development of an earned value cost/schedule system
- ❑ Indexes to monitor progress



The project control process

- ❑ Setting a baseline plan
- ❑ Measuring progress and performance
- ❑ Comparing plan against actual
- ❑ Taking actions

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- The baseline plan provides us with the elements for measuring performance.
 - The baseline is derived from the cost and duration information found in the WBS database and time-sequence data from the network and resource scheduling decisions.

- ❑ Setting a baseline plan
 - ❑ Measuring progress and performance
 - ❑ Comparing plan against actual
 - ❑ Taking actions
-
- Time and budgets are quantitative measures of performance.
 - Measurement of **time performance** is relatively easy and obvious. That is, is the critical path early, on schedule or late; is the slack of near-critical paths decreasing to cause new critical activities?
 - Measuring **performance against budget** (e.g. money, units in place, labour hours) is **more difficult** and is **not** simply a case of comparing actual versus budget.

The project control process

- ❑ Setting a baseline plan
- ❑ Measuring progress and performance
- ❑ Comparing plan against actual
- ❑ Taking actions



Earned value (EV)

- ❖ EV is necessary to provide a realistic estimate of performance against a time-phased budget.
- ❖ EV is defined as the budgeted cost of the work performed – **the percentage of the total budget** *actually* completed at a point in time.



The project control process

- ❑ Setting a baseline plan
- ❑ Measuring progress and performance
- ❑ Comparing plan against actual
- ❑ Taking actions

- Periodic monitoring and measuring the status of the project allow for comparisons of actual versus expected plans.
- The timing of status reports must be frequent enough to allow for early detection of variations from the plan and early correction of causes.
- Usually, status reports should take place every one to four weeks to be useful and allow for proactive correction.

Plans are seldom materialised as expected



"This plan will be much easier not to implement than the last plan we didn't implement."

The project control process

- ❑ Setting a baseline plan
- ❑ Measuring progress and performance
- ❑ Comparing plan against actual
- ❑ Taking actions

- If deviations from plans are significant, **corrective action will be needed** to bring the project back in line with the original or revised plan.
 - ❖ Conditions or scope can change
 - ❖ Require a change in the baseline plan to recognise new information



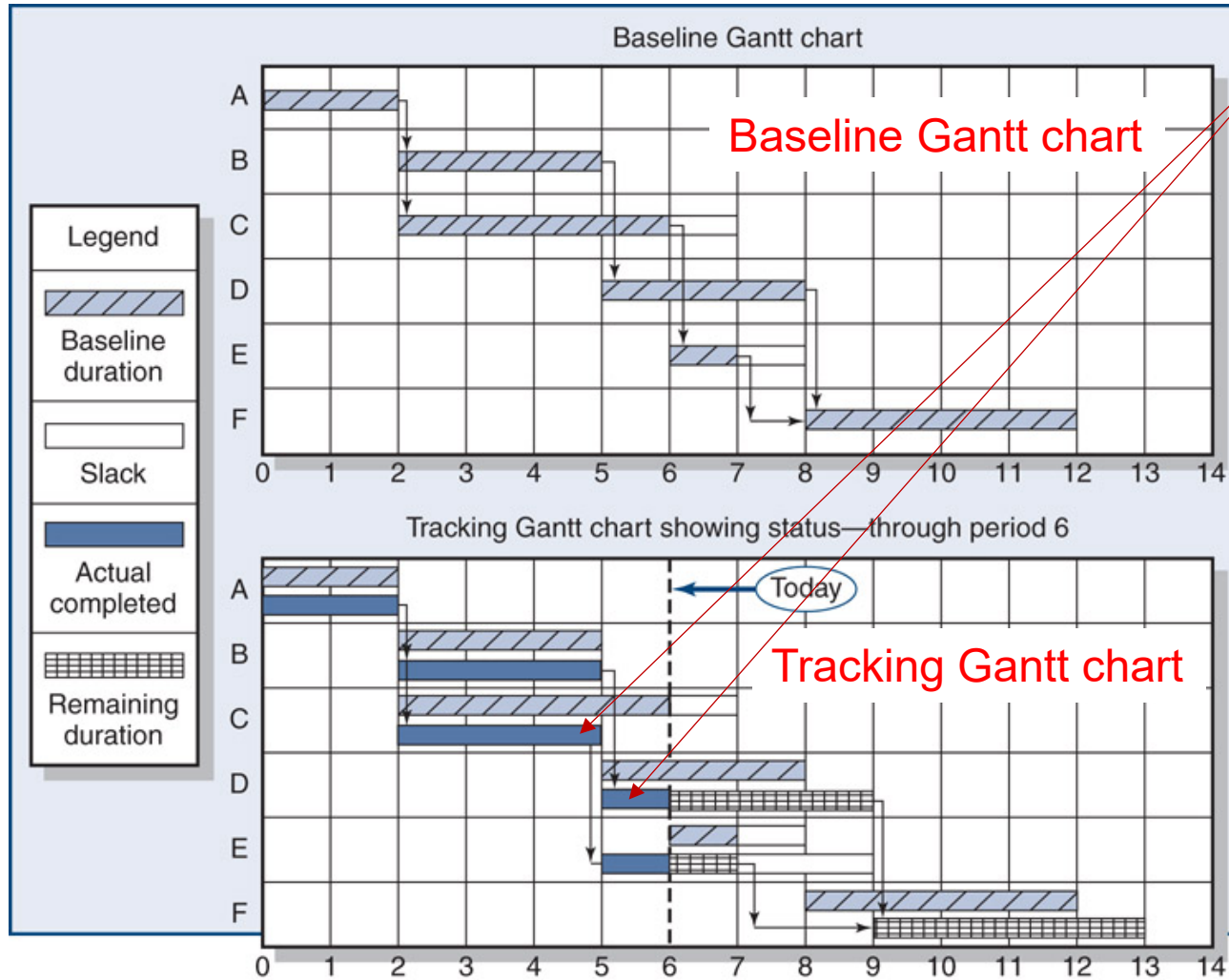
Monitoring the performance

To catch any negative variances from plan as early as possible to determine if corrective action is necessary.

- ❑ The project network schedule serves as the baseline to compare against actual performance
 - Tracking Gantt charts
 - Using Control charts

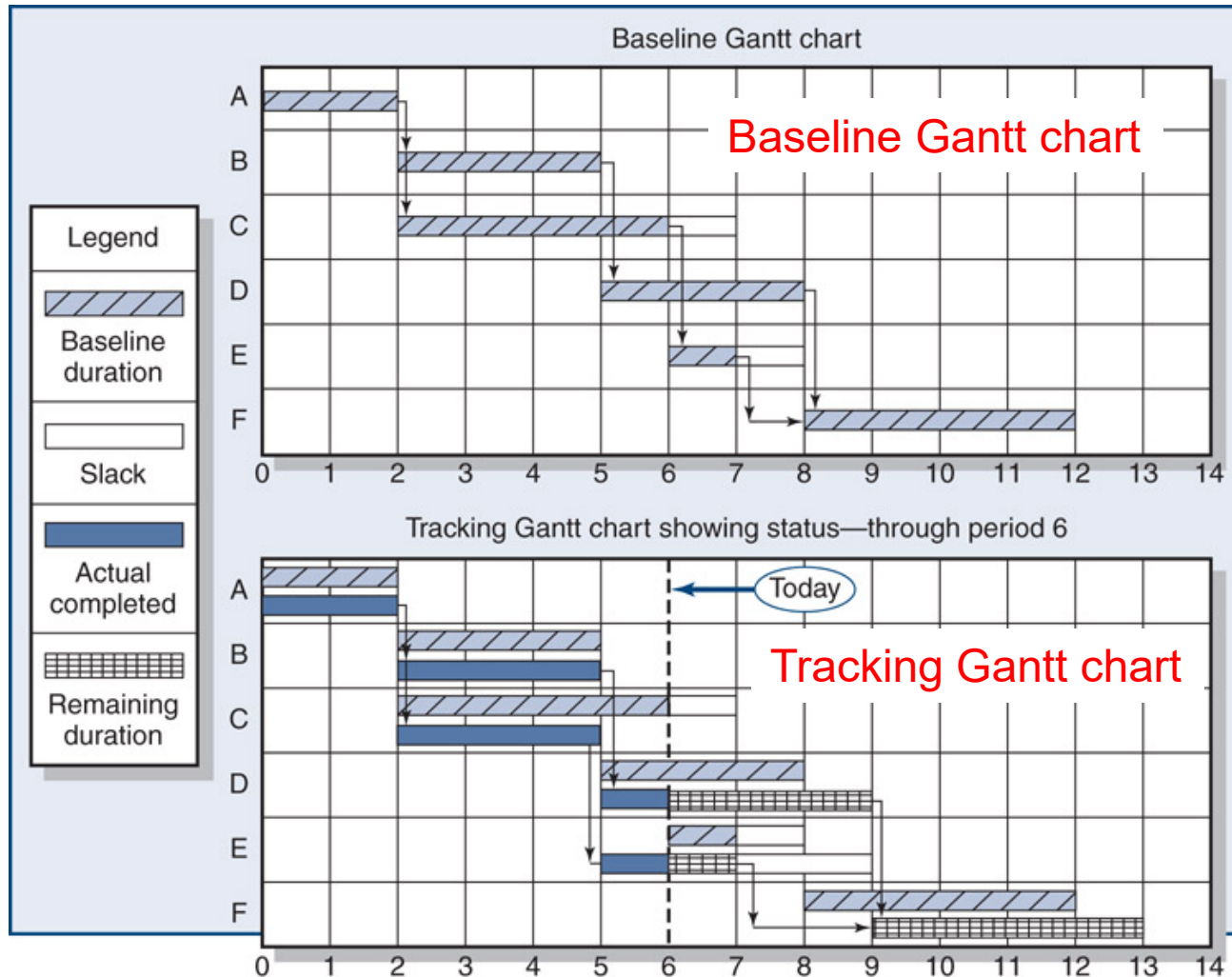


Tracking Gantt Chart



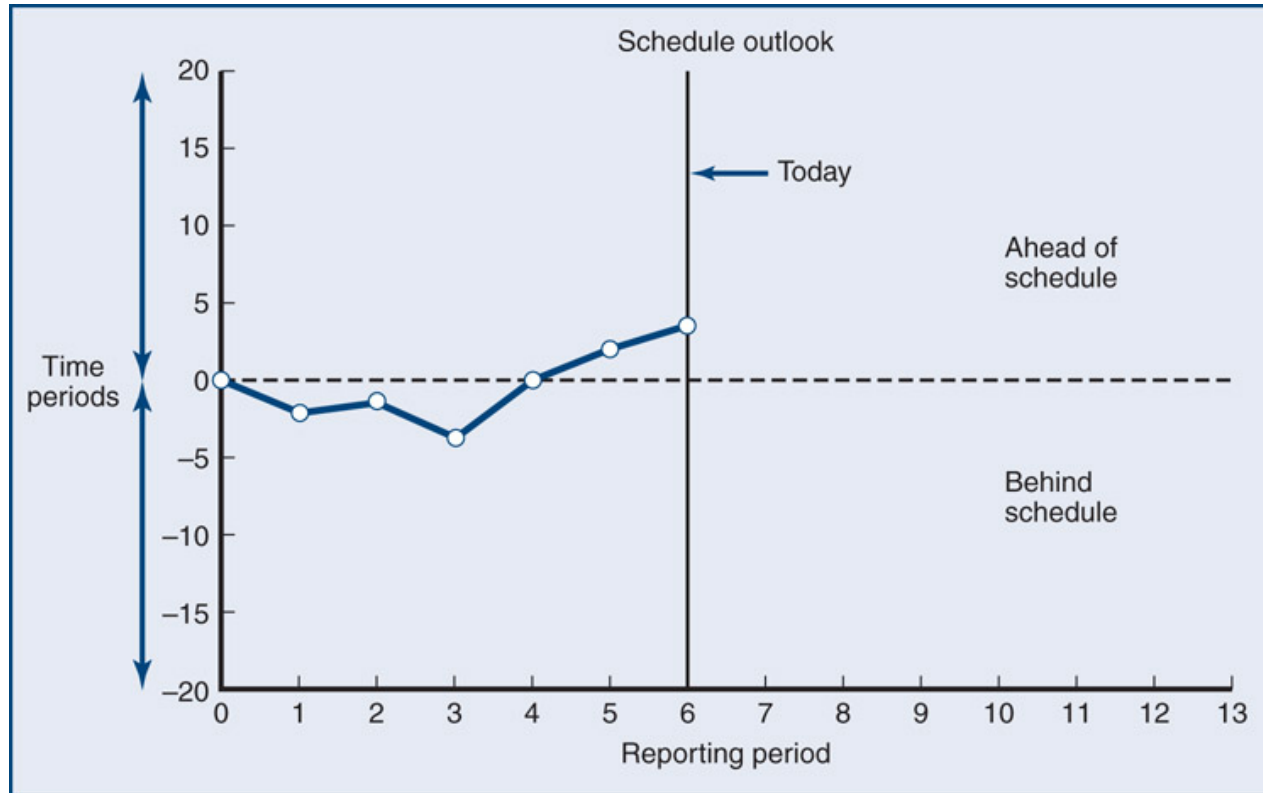
Solid bar - the actual start and finish times for completed activities or any portion of an activity completed

Tracking Gantt Chart



- The actual start time for activity C is period 2; the actual finish time is period 5;
- The actual duration is 3-time units, rather than 4 scheduled periods.
- Activities in process show the actual start time; the extended bar represents the expected remaining duration (see activities D and E).
- The remaining duration for activities D and E are shown with the hatched bar.
- Activity F, which has not started, shows a revised estimated actual start (9) and finish time (13).
- The project is now estimated to be completed one period later than planned.

Control Chart (Used to monitor milestone progress)

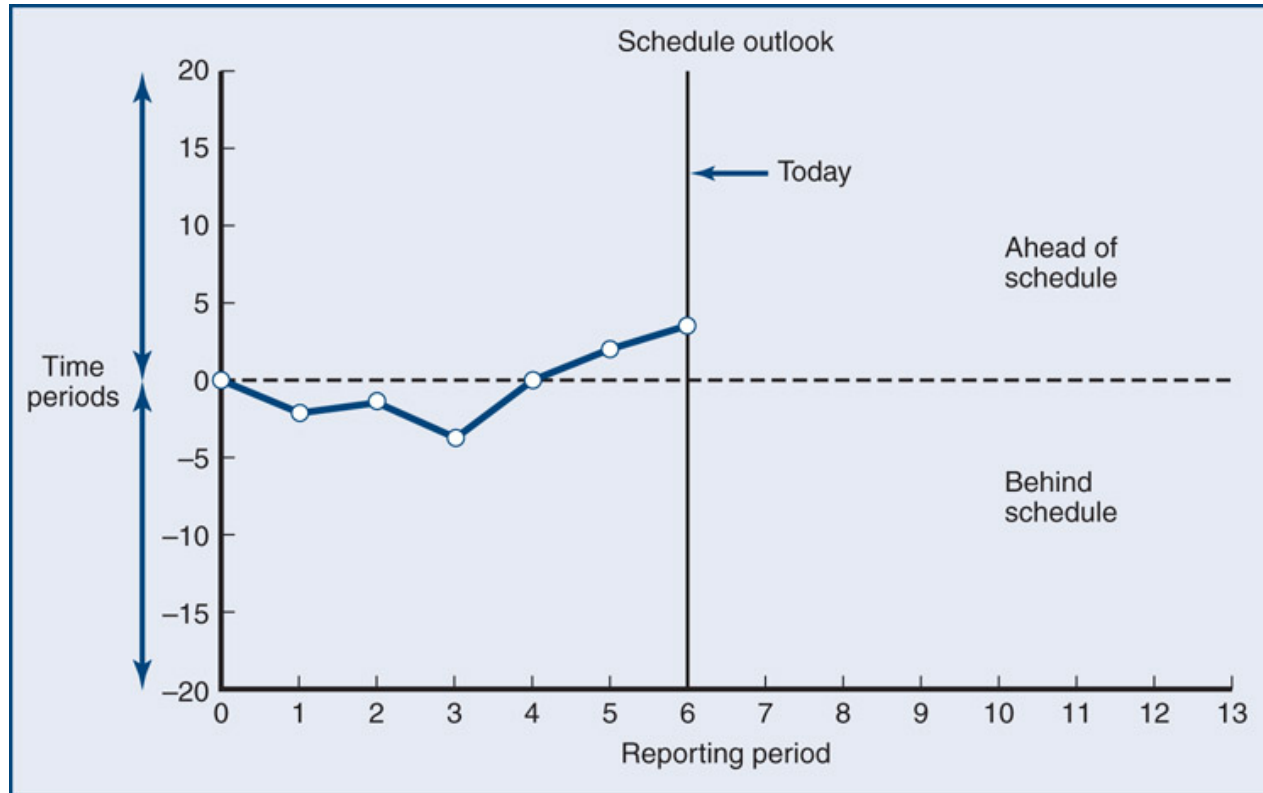


- Control charts are used to monitor progress towards milestones
- Milestones need to be concrete, specific, and measurable events
- Milestones must be easily identified by all project stakeholders – product testing complete
- Control charts are used to record and communicate project progress towards a milestone

Milestones – Significant project events that mark major accomplishment

Critical merge activities are good candidates for milestones

Control Chart



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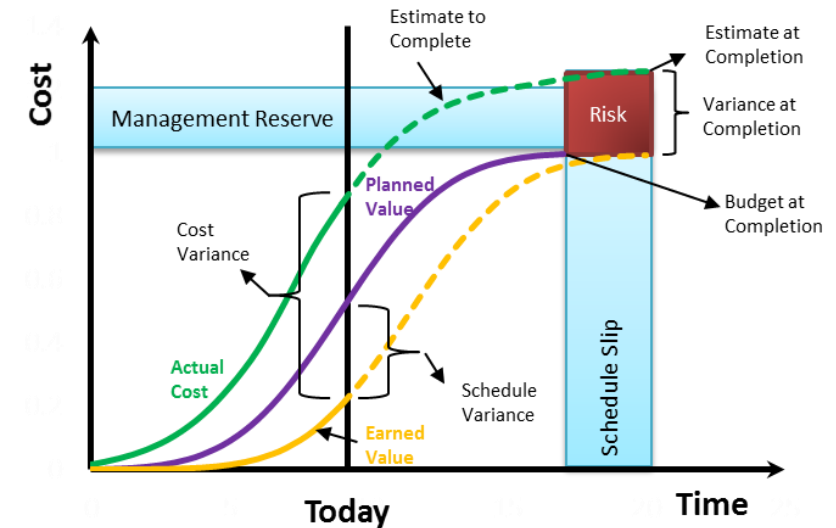
Milestones – Significant project events that mark major accomplishment

Critical merge activities are good candidates for milestones

Earned value cost/schedule system

Development of an earned value (EV) cost/schedule system

- ❑ The **EV system** starts with the time-phased costs that provide the project budget baseline - the **planned budgeted value** of the work scheduled (**PV**).
- ❑ Given this time-phased baseline, **comparisons are made with actual and planned schedules and costs using EV**.
- ❑ The EV approach provides the missing links not found in conventional cost-budget systems.
- ❑ At any point in time, a status report can be developed for the project.



Steps of integrating cost/schedule system

- **Step 1 - Define the work using a WBS.** This step involves developing documents that include the following information:
 - scope
 - work packages
 - deliverables
 - organisation units
 - resources
 - budgets for each work package
- **Step 2 - Develop work and resource schedule.**
 - Schedule resources to activities
 - Time-phase work packages into a network
- **Step 3 - Develop a time-phase budget using work packages included in an activity.**
 - The cumulative values of these budgets will become the baseline and will be called the planned budgeted cost of the work scheduled (PV).
 - The sum should equal the budgeted amounts for all the work packages in the cost accounts.

Steps of integrating cost/schedule system

- Step 4 - At the work package level, collect the actual costs for the work performed. These costs will be called the **actual cost** of the work completed (**AC**). Collect per cent complete and multiply this times the original budget amount for the value of the work completed. These values will be called **earned values (EV)**.
 - EV is the per cent of the original budget that has been earned by actual work completed.
- Step 5 - Compute
 - **Schedule variance** ($SV = EV - PV$) and
 - **Cost variance** ($CV = EV - AC$).

Prepare hierarchical status reports for each level of management—from work package manager to customer or project manager.

Steps of integrating cost/schedule system

- ☐ Schedule variance ($SV = EV - PV$)
- ☐ Cost variance ($CV = EV - AC$)
- **Cost variance** tells us if the work accomplished costs **more or less than was planned at any point over the life of the project.**
- If labour and materials have not been separated, cost variance should be reviewed carefully to isolate the cause to either labour or materials—or to both.
- **Schedule variance** presents an overall assessment of all work packages in the project scheduled to date. It is important to note schedule variance contains no critical path information. **Schedule variance measures progress in dollars rather than time units.**
- The only accurate method for determining the true time progress of the project is to compare the project network schedule against the actual network schedule to measure if the project is on time

Methods of variation analysis

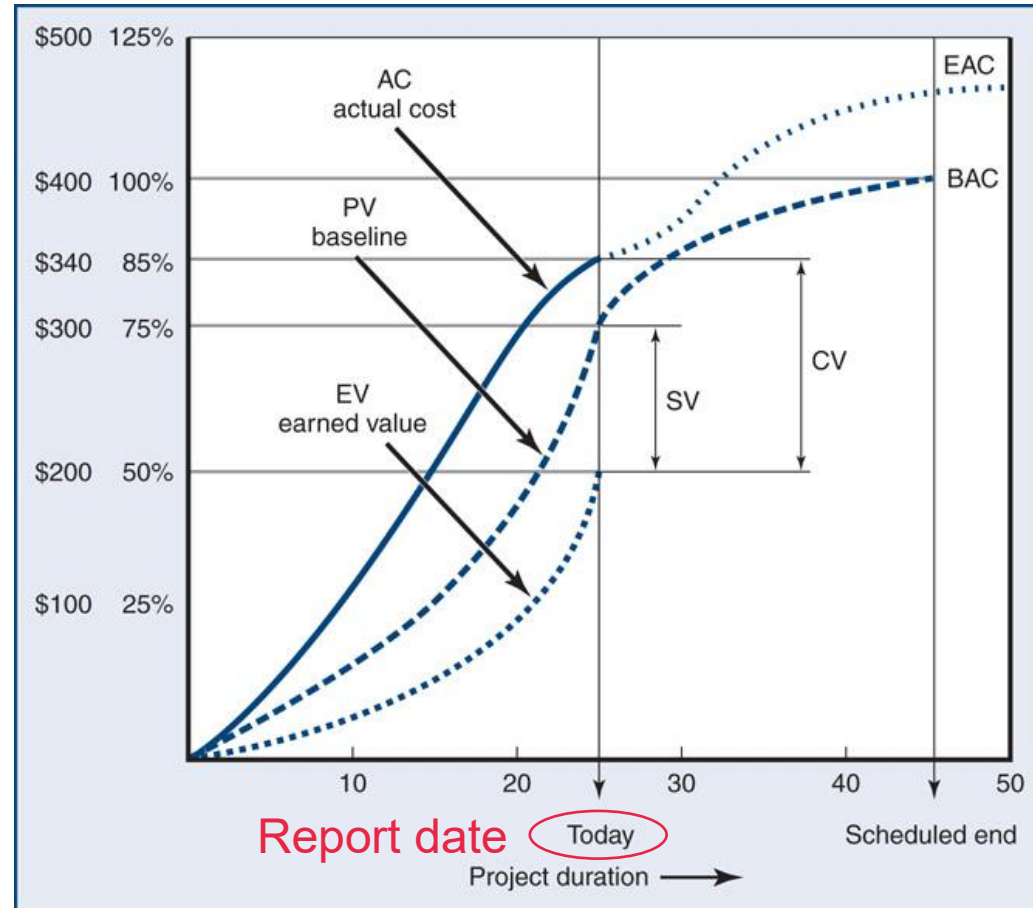
- ❑ Schedule variance ($SV = EV - PV$)
- ❑ Cost variance ($CV = EV - AC$)

The method for measuring accomplishments centres on two key computations:

- ❑ Comparing EV with the expected schedule value (Schedule variance)
 - ❑ Comparing EV with the actual costs (Cost variance)
-
- Planned cost of the work scheduled (PV)
 - Budgeted cost of the work completed (EV)
 - Actual cost of the work completed (AC)

Earned value cost/schedule system

Methods of variation analysis



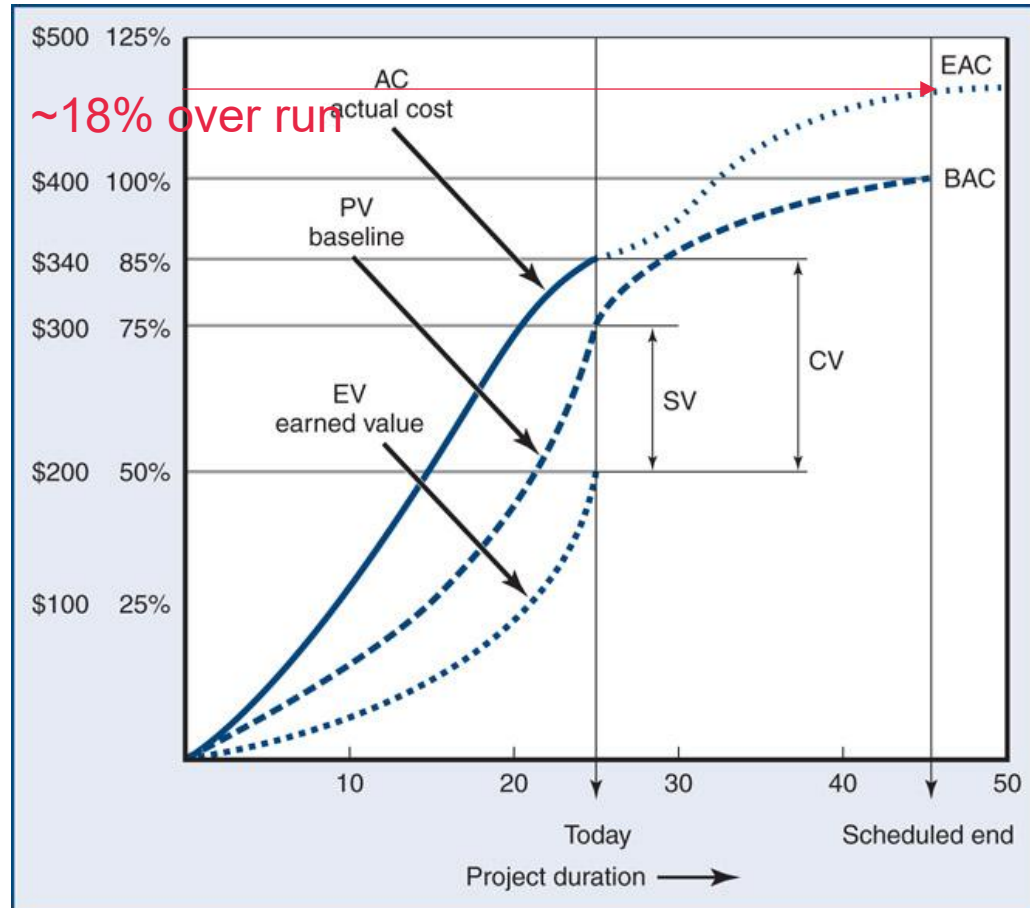
The graph also focuses on

- EV (Earned value)
- PV (Planned budgeted value)
- AC (Actual cost)
- what remains to be accomplished
- Any favourable or unfavourable trends.

A sample cost/schedule graph with variances identified for a project at the current status report date.

Earned value cost/schedule system

Methods of variation analysis



- At the end of period 25, 75 per cent of the work was scheduled to be accomplished. At the end of period 25, the value of the work accomplished is 50 per cent.
- The actual cost of the work completed to date is \$340, or 85 per cent of the total project budget. The graph suggests the project will have **about an 18 per cent cost overrun** and be five times units late.
- The current status of the project shows **the cost variance (CV) to be over budget by \$140** ($EV - AC = 200 - 340 = -140$).
- The **schedule variance (SV) is negative \$100** ($EV - PV = 200 - 300 = -100$), **which suggests the project is behind schedule**.

Performance Index

Project Per cent Complete Index

Performance Index

Project Per cent Complete Index

$$\text{Cost performance index (CPI)} = \frac{EV}{AC}$$

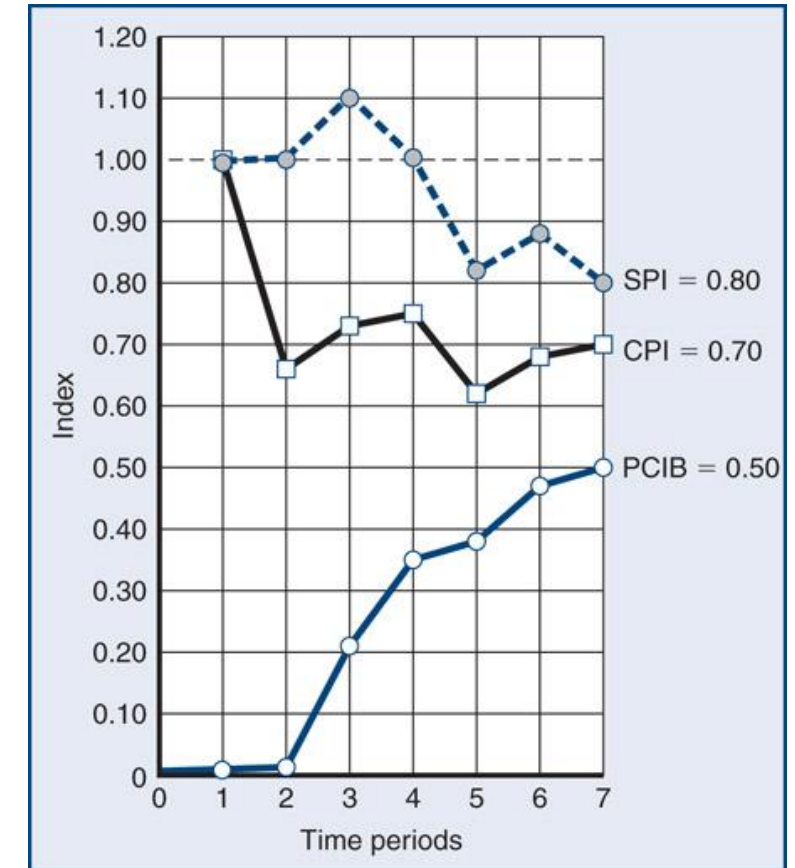
$$\text{Schedule performance index (SPI)} = \frac{EV}{PV}$$

If CPI is 0.7

\$0.70 worth of work planned to date has been completed for each \$1.00 actually spent—an unfavourable situation indeed. The CPI is the most accepted and used index. **It has been tested over time and found to be the most accurate, reliable and stable.**

If SPI is 0.8

The schedule index indicates \$0.80 worth of work has been accomplished for each \$1.00 worth of scheduled work to date.



Performance Index

Project Per cent Complete Index

$$\text{Per cent complete index (PCIB)} = \frac{EV}{BAC} = \frac{160}{320} = 0.5 \text{ (50\%)}$$

PCIB is 0.50 (50%)

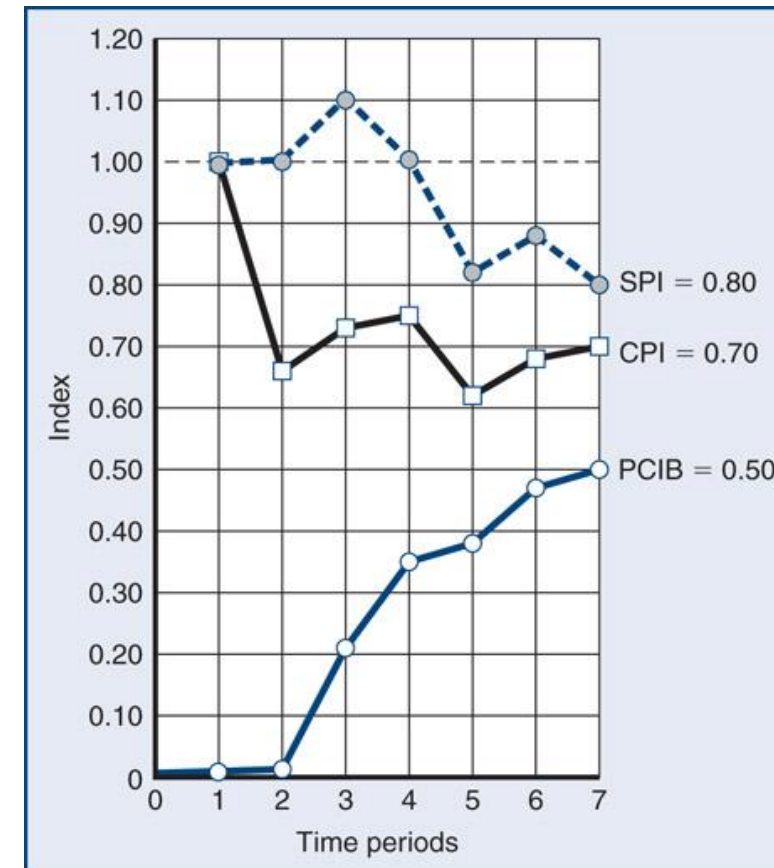
PCIB indicates the work accomplished represents 50 per cent of the **total budgeted** (BAC) dollars to date.

$$\text{Per cent complete index (PCIC)} = \frac{AC}{EAC} = \frac{230}{575} = 0.4 \text{ (40\%)}$$

PCIC is 0.4 (40%)

PCIC views per cent complete in terms of actual dollars spent to accomplish the work to date and **the actual expected dollars** for the completed project (EAC). For example, **at the end of period 7 the staff re-estimates that the EAC will be 575 instead of 320.**

BAC = \$320



Forecasting Final Project Cost

$$EAC_f = AC + ETC$$

$$ETC = \frac{\text{Work remaining}}{\text{CPI (Cumulative cost index to date)}} = \frac{BAC - EV}{EV/AC}$$

- Total baseline budget (BAC) for the project \$320
- Cumulative earned value (EV) to date \$160
- Cumulative actual cost (AC) to date \$230

$$EAC_f = AC + ETC = 230 + \frac{320 - 160}{\frac{160}{230}} = 230 + \frac{160}{0.7} = 230 + 229 = 459$$

$$BAC = \$320$$

$$EAC_f = \$459$$



THANK YOU

Next Lecture – Guest Lecture (BB – For Announcement)